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Practice Scenarios for Use Case Diagram and Sequence Diagram

This document contains **10 detailed, exam-oriented scenarios** designed specifically for **practicing UML Use Case Diagrams and Sequence Diagrams**.

Each scenario is intentionally written in **problem-statement style**, similar to what students see in **midterms/finals**, without giving diagrams directly.

For each scenario, students are expected to: - Identify **actors** - Draw a **Use Case Diagram** - Draw **Sequence Diagrams** - Handle **main flow, alternative flow, and exception flow**

Scenario 1: University Course Registration System

A university has developed an online course registration system for students. A student must log in using their university ID and password. After successful login, the student can view a list of offered courses for the semester. When the student selects a course, the system checks whether: - the student has completed the required prerequisites - seats are available in the course

If both conditions are satisfied, the course is added to the student's registered courses and a confirmation message is shown. If prerequisites are missing or seats are full, the system displays an appropriate error message.

Additional Conditions: - A student may drop a course before the registration deadline. - If the system fails while registering, the transaction must be rolled back.

Scenario 2: ATM Cash Withdrawal System

A bank customer uses an ATM machine to withdraw cash. The customer inserts the ATM card and enters a PIN. The system validates the PIN with the bank server. After successful authentication, the customer selects an account type and enters the withdrawal amount.

The system verifies whether sufficient balance is available. If yes, cash is dispensed and the account balance is updated. Otherwise, an error message is displayed.

Additional Conditions: - If the PIN is entered incorrectly three times, the card is blocked. - If the ATM runs out of cash during the transaction, the withdrawal is cancelled.

Scenario 3: Online Shopping and Payment System

An online shopping website allows customers to purchase products. A customer browses products, adds items to the shopping cart, and proceeds to checkout. At checkout, the customer enters shipping details and selects a payment method.

The system sends payment details to an external payment gateway. If payment is successful, an order is created and confirmation is sent to the customer. If payment fails, the order is not created.

Additional Conditions: - A customer can update or remove items from the cart before checkout.
- Payment gateway timeout should be handled gracefully.

Scenario 4: Library Book Issuing System

A university library uses a computerized system to issue books. A library member presents their membership ID and requests a book. The librarian searches the book in the system.

If the book is available and the member has not exceeded the borrowing limit, the book is issued and the due date is recorded. Otherwise, the system displays a rejection message.

Additional Conditions: - A member may reserve a book if it is currently unavailable. - If the system is offline, manual issuing is not allowed.

Scenario 5: Online Examination System

Students take exams through an online examination system. A student logs in at the scheduled exam time. After authentication, the system displays the exam instructions and starts the timer.

The student answers multiple-choice questions and submits the exam. The system automatically evaluates the answers and stores the result.

Additional Conditions: - If the internet connection is lost temporarily, answers should be auto-saved. - If time expires, the exam is automatically submitted.

Scenario 6: Food Delivery Mobile Application

A food delivery application allows customers to order food from nearby restaurants. A customer selects a restaurant, chooses food items, and places an order. The system sends the order to the restaurant for preparation.

Once the food is ready, a delivery rider is assigned. The customer can track the order until delivery.

Additional Conditions: - If no rider is available, the order is delayed. - A customer may cancel the order before preparation starts.

Scenario 7: Hospital Appointment Booking System

A hospital provides an online appointment booking system. A patient searches for doctors by specialization and views available time slots. The patient selects a slot and confirms the appointment.

The system updates the doctor's schedule and sends confirmation to the patient.

Additional Conditions: - If the selected slot becomes unavailable, the system asks the patient to choose another slot. - Patients can cancel appointments before the appointment time.

Scenario 8: Railway Ticket Reservation System

Passengers can book train tickets online. A passenger searches for trains by entering source, destination, and travel date. The system displays available trains and seat status.

After selecting a train and seat, the passenger makes payment. On successful payment, the ticket is confirmed and generated.

Additional Conditions: - If seats are not available, the passenger may be placed on a waiting list. - Payment failure should not reserve the seat.

Scenario 9: Student Attendance Management System

Teachers mark student attendance using an online system. A teacher logs in, selects a course, and views the list of enrolled students. The teacher marks students as present or absent and submits the attendance.

The system saves the attendance record and updates student attendance percentages.

Additional Conditions: - Attendance cannot be edited after submission. - If the session expires, unsaved data is lost.

Scenario 10: Ride-Hailing Application

A ride-hailing mobile application allows users to book rides. A passenger enters pickup and drop-off locations and requests a ride. The system finds nearby drivers and sends ride requests.

When a driver accepts the request, the ride is confirmed and tracking begins. After ride completion, payment is processed automatically.

Additional Conditions: - If no driver accepts the request, the ride is cancelled. - If payment fails after ride completion, the user is notified.